

NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM II

MH2100 – Calculus III

November 2018

TIME ALLOWED: 90 minutes

Name:

Matric. no.:

Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. A two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1.**(40 marks)**

Circle (only) the correct answer for each question. Each question is worth 5 marks.

- (i) Calculate the double integral $\iint_R (6x^2y^3 - 10x^4) dA$ where $R = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 4\}$.

- (a) 190
- (b) 170
- (c) 130
- (d) 150
- (e) 120

- (ii) Use polar coordinates to find the volume of the solid under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$ (in the plane $z = 0$).

- (a) 81π
- (b) 162π
- (c) 324π
- (d) 27π
- (e) None of the above

- (iii) Evaluate the iterated integral by converting to polar coordinates.

$$\int_{-5}^5 \int_0^{\sqrt{25-y^2}} (x^2 + y^2)^{3/2} dx dy$$

- (a) 625π
- (b) 1250π
- (c) $\frac{625\pi}{2}$
- (d) $\frac{625\pi}{4}$
- (e) None of the above

Questions continue on the next page.

(iv) Evaluate the iterated integral.

$$\int_0^\pi \int_0^1 \int_0^{\sqrt{1-y^2}} 4y \sin x dz dy dx$$

- (a) $\frac{8}{3}$
- (b) 4
- (c) $\frac{4}{3}$
- (d) $\frac{2}{3}$
- (e) None of the above

(v) Use cylindrical coordinates to evaluate $\iiint_T \sqrt{x^2 + y^2} dV$, where T is the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 4$ and $z = 8$.

- (a) $\frac{8\pi}{3}$
- (b) 4π
- (c) $\frac{4\pi}{3}$
- (d) $\frac{2\pi}{3}$
- (e) None of the above

Questions continue on the next page.

- (vi) Use spherical coordinates to evaluate $\iiint_B (x^2 + y^2 + z^2)^2 dV$ where B is the ball with center the origin and radius 4.

- (a) $\frac{65536\pi}{7}$
- (b) $\frac{4096\pi}{5}$
- (c) $\frac{8192\pi}{3}$
- (d) $\frac{32768\pi}{7}$
- (e) None of the above

- (vii) Find the Jacobian of the plane transformation given by $x = 5u \sin v$ and $y = 4u \cos v$.

- (a) $\frac{\partial(x, y)}{\partial(u, v)} = 9u$
- (b) $\frac{\partial(x, y)}{\partial(u, v)} = 36u$
- (c) $\frac{\partial(x, y)}{\partial(u, v)} = -20u$
- (d) $\frac{\partial(x, y)}{\partial(u, v)} = -u$
- (e) None of the above

- (viii) Evaluate the iterated integral.

$$\int_1^3 \int_y^3 12xy \, dx dy$$

- (a) 96
- (b) 116
- (c) 106
- (d) 126
- (e) None of the above

QUESTION 2.**(30 marks)**

- (a) Evaluate the iterated integral

$$\int_0^4 \int_{\sqrt{y}}^2 e^{x^3} dx dy.$$

- (b) Use the transformation $x = \sqrt{2}u - \sqrt{\frac{2}{3}}v$; $y = \sqrt{2}u + \sqrt{\frac{2}{3}}v$ to evaluate the integral

$$\iint_R x^2 - xy + y^2 dA,$$

where R is the region bounded by the ellipse $x^2 - xy + y^2 = 2$.

- (c) Evaluate the triple integral

$$\iiint_R z^2 dV,$$

where R is the solid bounded by the sphere $x^2 + y^2 + z^2 = 1$, the cylinder $x^2 + y^2 = 1$, and the plane $z = 2$.

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QUESTION 3.**(30 marks)**

- (a) Find the work done by the force field $\mathbf{F}(x, y) = x \sin(y)\mathbf{i} + y\mathbf{j}$ on a particle that moves along the curve $y = x^2$ from $(0, 0)$ to $(\sqrt{\pi}, \pi)$.
- (b) Evaluate the line integral $6 \int_C yz \cos x ds$ where C has parametrisation $x = t$, $y = \cos t$ and $z = \sin t$, with $0 \leq t \leq \pi$.
- (c) Let $\mathbf{F}(x, y) = x^3 y^2 \mathbf{i} + y(\frac{x^4}{2} - 2)\mathbf{j}$ and let C be a curve with parametrisation $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + (1 + t^3)\mathbf{j}$ with $0 \leq t \leq 1$ (starting at $\mathbf{r}(0)$ and terminating at $\mathbf{r}(1)$).
- (i) Find a function f such that $\mathbf{F} = \nabla f$.
- (ii) Hence, or otherwise, evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

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END OF PAPER